Supplementary file 1 — GRRAS checklist

Guidelines for Reporting Reliability and Agreement Studies (Kottner et al., 2011)

Manuscript: *What can be measured from the broadcast? Validity of single-camera video for football-kick kinematics and swing-leg joint loads.* Section numbers refer to the submitted manuscript.

#	GRRAS item	Where it is reported
1	Identify in title or abstract that interrater/intrarater reliability or agreement was investigated	Title (“validity”); abstract (“method-comparison study”, agreement against motion capture)
2	Name and describe the diagnostic or measurement device of interest explicitly	§1 (broadcast-video pipeline); §2.3 (RTMW-x detector, MotionAGFormer-B lifter, scaling, alignment, contact)
3	Specify the subject population of interest	§1 (football players kicking in broadcast matches); §2.1 (laboratory participants as reference population and its limits)
4	Specify the rater population of interest (if applicable)	Not applicable: the method is automated (detector + lifter). The reference “rater” is optical motion capture (§2.1)
5	Describe what is already known about reliability and agreement and provide a rationale for the study	§1
6	Explain how the sample size was chosen; state the number of raters, subjects/objects and replicate observations	§2.7 (fixed by the public datasets; 83 participants); §2.1 (7 CMU kicks); §2.3 (5 views × 5 sizes × 2 frame rates; 3 noise seeds for CMU, 1 for MoVi)
7	Describe the sampling method	§2.1 (all CMU kick trials; all MoVi participants with a usable, gap-free kick; all 200 3DSP shots)
8	Describe the measurement process	§2.2 (features, loads, windows); §2.3 (virtual camera, error models, lifting); §2.4 (real video, digital twin); §2.5 (remedies)
9	State whether measurements were conducted independently	§2.3: video estimates never used the reference except for scale (thigh length) and one rigid alignment per sequence, which emulate known stature and pitch calibration; the adapted lifter was tested only on unseen participants and held-out broadcast residuals (§2.5)
10	Describe the statistical analysis	§2.7 (bias, MAE, limits of agreement, ICC(2,1), calibration slope, bootstrap CIs, acceptability criteria, TOST, SPM); §2.8 (injury-study size)
11	State the actual number of raters and subjects included and the number of replicate observations	§2.1 and Table 1; §3.4 (82 participants × 2 cameras); Tables 2–4 (n per analysis)
12	Describe the sample characteristics of raters and subjects	§2.1 (CMU: 2 participants; MoVi: 83 participants, 55 female, non-footballers, air kicks)
13	Report estimates of reliability and agreement including measures of statistical uncertainty	§3.1–3.7, Tables 2–4 (bias, MAE with 95% CI, 95% LoA, ICC with 95% CI); Supplementary Figures
15	Provide detailed results if possible (e.g. online)	Supplementary file 2 (complete statistical outputs), code and aggregated results in the public repository